Aegis Station

Gradient One

Three-Arm Multi-Gravity Validation Platform, 1.6 rpm

Design Dossier | Rev C — November 2026 (Rev B, September 2026, with the crossing taken in) | Not for Hardware Commitment

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1. Purpose

Gradient One (G1) is a rotating orbital research platform that places Earth gravity, Mars gravity, lunar gravity and a rotating microgravity hub in one vehicle at one spin rate, for six-month crewed campaigns. It is the platform defined to close the Artificial Gravity Requirements Evaluation (AGRE) and to retire the rotating-structure risks that Aegis Station carries as assumptions: 1g at 1.6 rpm for months, the hub/rotor boundary under rotation, radial fluids at full head, balance authority, and spin transients with crew aboard.

Two questions, one vehicle. The station's question is whether people can live and work for months at 1g produced by rotation at 1.6 rpm. The surface program's question is how bone, cardiovascular, ocular and plant endpoints respond to lunar and Mars gravity over months, against a 1g control in the same vehicle. Both need several gravity levels, held for a long time, in one place, with rotation present.

> **Rev B in one paragraph.** The single-arm truss with a counterweight becomes a self-balancing three-arm rotor that spins about its major axis. The despun hub becomes a small docking node joined to a rotating 0g hub by a cyclic transfer lock, contactless power and an optical data link, with no continuous rotary seal in any life-critical path. Campaigns move from 30–90 days to 180 days, crew from 2–4 to 10, because the requirements ask for six-month exposures. Mass is bottom-up at about 227 t. The cost class moves from $0.6–1.4 B to $5–10 B to first campaign.

> **Rev C in one paragraph.** Rev B with its one unspecified element specified. The transfer lock, carried as a 2 t placeholder, becomes the station's crossing scaled to Gradient One: a drum of three two-crew chambers between the rotating hub's upper face and the despun node's lower face, locked to one frame or the other and spun between them by its own drive, 13.8 t dry with its chambers and mechanisms. The hub's dimensions are fixed with it. The vehicle grows to about 242 t on orbit; nothing else in the sizing moves. The crossing is in the dynamics model as a body with its own spin and two locks, and the flight software flies it end to end; the figures are on the Gradient One page.

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2. Geometry and gravity bands

Spin rate is fixed by the 1g radius: $\omega = \sqrt{(g/R)} = 0.167$ rad/s = 1.60 rpm at R = 350 m. Band radii follow from $a = \omega^2 r$. Each module is mounted radially with three decks at 2.4 m pitch; the band radius is the middle deck.

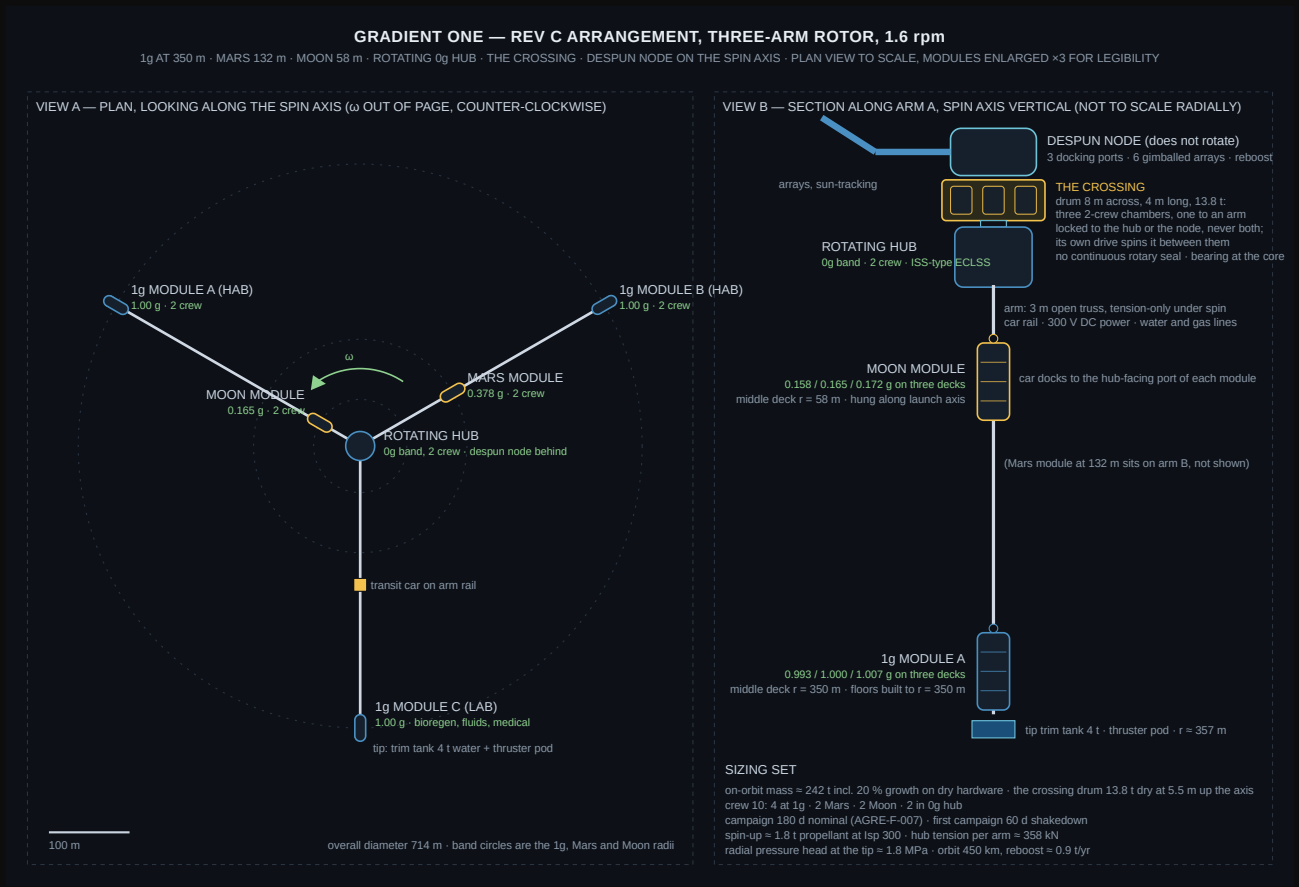
Band	Middle deck radius	Gravity, lower / middle / upper deck	Location	Crew
1g	350.0 m	1.007 / 1.000 / 0.993 g	Three tip modules: habitats A and B, laboratory C	4
Mars	132.4 m	0.385 / 0.378 / 0.371 g	Inboard on arm B, cradle-mounted	2
Moon	57.8 m	0.172 / 0.165 / 0.158 g	Inboard on arm A, cradle-mounted	2
0g, rotating	0 m	< 0.001 g; 1.6 rpm present	Rotating hub module	2
0g, non-rotating	on axis	0 g; no rotation	Despun node, transit only	–

Kinematic quantity	Value
Angular rate	0.1674 rad/s, 1.60 rpm, period 37.5 s
Coriolis at 1 m/s in the spin plane	0.335 m/s², 3.4 % of g, either band
Head-to-foot gradient over 2 m at 1g	0.57 %
Tip radius (module bottom plus trim tank)	357 m; overall diameter 714 m
Floor curvature, 1g modules	Floors built to r = 350 m, as the station's rings are

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3. Configuration

A Y-shaped rotor: three open-truss arms at 120°, each about 344 m from hub port to tip, with an identical habitat module hung from each tip at the 1g radius. The lunar module sits inboard on arm A and the Mars module inboard on arm B. A rotating hub module joins the arms and houses the 0g crew: a 4.5 m node-class hull 6 m long inside the 12 m ring the arms hinge on, its upper face 3 m up the axis. A despun node on the spin axis, its lower face at 8 m, carries three docking ports, six deployable arrays on two gimbals, reboost thrusters and antennas. Between the two faces sits the crossing (section 5): an 8 m drum, 4 m long, on a 3 m bearing, its three chambers at 2.5 m from the axis, one to each arm's port.



Why three arms

A planar rotor with three or more equal arms has spin-axis inertia twice its transverse inertia, so it spins about its major axis and energy dissipation from crew, fluids and flexure drives it toward the spin it already has. A two-arm dumbbell has equal spin and in-plane transverse inertia and no such preference. Three identical tip modules also balance in the plane without a dead counterweight, which removes the Rev A counterweight and its ambiguity.

Why 350 m and 1.6 rpm

Earth is the 1g control for the biology; the radius is chosen because 1g at 1.6 rpm is the station's operating point and AGRE reports tolerance per spin rate. The sizing model quantifies the alternative:

Parameter	Rev C, 1.6 rpm	Alternative, 3 rpm
1g / Mars / Moon radius	350 / 132 / 58 m	99 / 38 / 16 m
Coriolis at 1 m/s	3.4 % g	6.4 % g
On-orbit mass	~242 t	~209 t
Saving	–	~32 t, ~13 %, arms only; modules, boundary, assembly and operations unchanged
Measures the station's column of the AGRE table	Yes	No

Radius is cheap on this vehicle; crew-rated volume is what costs. The 1g band earns its 350 m because the arm to reach it is a few tonnes of truss and the question it answers is the station's.

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4. Modules

One common pressurized module, five copies: 4.5 m diameter, 11 m hull, three 2.4 m decks and a utility deck, about 14 t dry and 17 t outfitted, in the class of the ISS laboratory and node modules. Every module is mounted radially, long axis along the arm, so the centripetal load runs through the hull along the axis its launch structure already carries at several g. Tip modules hang from the arm end; the Moon and Mars modules sit beside the arm on cradles carrying 0.17 g and 0.38 g of their mass, 28 kN and 63 kN respectively. The hub-facing end of every module carries the port that the transit car docks to.

Module	Role	Notes
1g habitats A and B	Crew quarters, galley, exercise, hygiene; vehicle water recovery and oxygen generation	Water processing at 1g uses gravity phase separation and is itself a validation item
1g laboratory C	Bioregenerative growth racks; radial fluids and two-phase test bed; medical and imaging suite for all bands	Same hull as A and B
Mars and Moon modules	Two crew each; growth racks; physiology instrumentation	Atmosphere revitalization local; water processed at 1g and returned by the arm lines
Rotating hub	Node-class, 4.5 m across and 6 m long inside the arm-root ring; three radial arm ports, one axial port to the crossing on its upper face; two crew at 0g with ISS-type microgravity life support; rotor contingency refuge	14 t
The crossing	A drum of three two-crew chambers between the hub and the node, locked to one frame or the other, spun between them by its own drive; the chambers push a metre to the ports and mate; hatches on both faces	13.8 t dry: 7 t drum, three 1.6 t chambers, 2 t of locks, berths and drive
Despun node	Three docking ports, arrays, battery, reboost, communications; pressurized for transit and cargo handling only	8 t pressurized, plus boundary and power hardware

Three decks give a measured gravity gradient between floors (AGRE-F-004) and radial stair and ladder transit (AGRE-F-002) inside every module, without the pressurized spoke a full-scale habitat would need.

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5. Hub/rotor boundary

Rev B adopted the principle of the station's hub/ring boundary: no continuous rotary mechanical interface carries a life-critical flow. Every crossing is cyclic, contactless, or eliminated by distributing the function to both sides.

Flow	Crossing	Heritage
Crew and cargo	The crossing (Rev C): the station's crossing at three chambers. The drum idles locked to the rotor with its chambers at the arms' ports; to cross, the chambers retract and seal, the rotor lock releases, the drum's drive brings it to rest against the node, the node lock engages within 10 mm at the berths, the chambers push a metre to the node's ports and mate, and the hatches open only with the tunnel equalized. No seal slips under pressure. The drive reacts into the rotor.	Hatches and docking mechanisms; the station's crossing at the station's rate with a seventieth of its inertia
Power	Contactless rotary transformer across the bearing; arrays and battery on the despun side	Rotary transformers in flight mechanisms
Data	Free-space optical link on the spin axis	Optical intersatellite links
Heat	Rejected on each side independently; no fluid crosses	Body and panel radiators
Thrust and torque	Reboost thrust from the despun node passes through the bearing as a radial load; a bearing drive holds the node despun	Dual-spin spacecraft

The vehicle is a dual-spin spacecraft: a rotor spinning about its major axis and a despun platform on the same bearing, an arrangement flown for decades on communications satellites at far higher rates. A seized bearing turns the node into a co-rotating hub; the two crew vehicles that remain docked for the whole campaign are the crew's way home in either case. The boundary's performance under rotation is a campaign objective (AGRE-F-012) at the station's spin rate.

The crossing, sized

Quantity	Value	Basis
Drum	8.0 m outer diameter, 3.0 m bore over a 2.5 m core, 4.0 m long	Option C of the sizing, 2026-11-07: the hub's dimensions as they are
Chambers	Three, 2.0 m across, two crew each, at 2.5 m from the axis	One to each arm's port
Where it sits	Mass centre 5.5 m up the axis, between the hub's face at 3 m and the node's at 8 m	
Mass	13.8 t dry; 16.6 t with growth	Replaces the 2 t transfer-lock placeholder
Speed and gravity at a berth	0.42 m/s; 0.007 g in a chamber at rate	The station's 1.41 m/s
Drum inertia, crew aboard	1.1×10^{5} kg·m²	A seventieth of the station's passenger drum
Angular momentum at rate	1.9×10^{4} N·m·s, 1 in 140,000 of the rotor's	
Drive torque, crew / cargo	156 / 183 N·m over 120 s; peak power 26 W	Into the node it would take 37 % of the 500 N·m despin drive, so the drive reacts into the rotor
Lock tolerance	10 mm at the berths (0.23°); 3×10^{-4} rad/s on rate	A third of the station's on rate: the engagement kicks the node
Push stroke	1.0 m; soft capture at 0.9 m, hard mate at 1.0 m	
Throughput	Six crew a cycle, thirty an hour; the whole crew in two cycles	
What a crossing costs the rotor	4 parts in a million of its spin; nothing to the despin drive but the bearing's friction	Dynamics model, section 9 of its report
What the engagement costs the node	2×10^{-5} rad/s at Gradient One's tolerance, a fifth of the lost threshold	Dynamics model

6. Dynamics, loads and balance

Quantity	Value	Basis
Spin-axis inertia	1.1×10^{10} kg·m²	Point masses at band radii plus distributed arms
Inertia ratio, spin to transverse	2.0	Planar three-arm rotor
Angular momentum	1.8×10^{9} N·m·s	$I\,\omega$
Rotational kinetic energy	43 kWh	$\frac{1}{2} I\,\omega^2$
Spin-up propellant	~1.8 t	Tip thruster pods at 357 m, Isp 300 s
Tension at hub, most loaded arm	~360 kN	Tip module, trim tank, car, arm self-load, Mars module
Equivalent aluminium section	18 cm² at 200 MPa	Arm sized by stiffness and rail, not strength
CM shift, one person 1g to hub	0.15 m	100 kg × 350 m / 242 t
CM shift, one car tip to hub	2.9 m	2 t × 350 m / 242 t; trimmed in advance, and carried by the car's balance weight
Trim authority	Three 4 t tip tanks, water pumped between them	Also the vehicle water reserve
Car Coriolis load at 2 m/s	~1.3 kN	$2\,m\,\omega\,v$

Attitude

The spin axis is held near the orbit normal so that the reboost direction stays in the plane the despun node can point in. For a flat rotor the orbit-averaged gravity-gradient torque precesses the spin axis about the orbit normal at $\Omega = \frac{3}{4}\, n^2 \cos\theta / \omega$, about 28° per day at 450 km. The orbit normal itself regresses at 4.9° per day at 51.6°. The axis therefore follows the orbit with a lag of order 10°, without propellant. The coupled motion is confirmed in the dynamics model (November 2026): the axis rides about 9° off the orbit normal and keeps station with the orbit's plane for nothing; the despun node's pointing and array shadowing remain open.

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7. Radial transit and radial plumbing

Crew move between bands in a pressurized two-seat car on a rail along each arm, docking to the hub-facing port of each module. A pressurized spoke tube, which is what the station will need, would add about 60 t at roughly 90 kg per metre over three arms and is deferred to the station's own design. The

car gives the inter-band transition measurements (AGRE-F-005, F-006) and its transit is a scheduled, trimmed event.

The water and gas lines along the arms are the rotating-frame fluids test bed. Static head from hub to tip is $\rho\omega^2 r^2/2$ = 1.8 MPa, about 18 atmospheres; the station's rings at 390 m see 1.9 MPa. Pumps at the tip work against it; a line draining toward the hub has it for free. Coriolis asymmetry in radial flow, two-phase behaviour and separator performance at each gravity level are measured on the lines and in the laboratory module (AGRE-F-011).

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8. Systems

System	Rev C sizing
Orbit	450 km, 51.6°. Effective drag area ~900 m² (modules, arrays, trusses at 15 % fill, node); drag ~0.09 N at mean solar density; Δv ~12 m/s per year; reboost ~0.9 t per year at Isp 300 s from the despun node.
Power	Load ~55 kW: five habitat modules at 5 kW; hub and node 6 kW; growth racks, 6 m² per band at 0.6 kW/m², 11 kW; exercise, medical and science 8 kW; arm utilities, cars, trim pumps and heaters 5 kW. Six 20 kW deployable arrays on two gimbals: ~102 kW sunlit at end of life, ~58 kW orbit-average. Battery 82 kWh for a 36-minute eclipse at 40 % depth of discharge. Arm distribution at 300 V DC, ~1 t of copper.
Thermal	Rejected per module by body-mounted and arm-mounted panels, ~0.5 t per module. No long fluid loops; nothing crosses the boundary.
Life support	Atmosphere revitalization local to each module. Water recovery and oxygen generation centralized in the 1g habitats. The hub keeps ISS-type microgravity equipment. Consumables ~7 t per 180-day campaign for ten crew at ~4 kg per crew-day, two to three cargo flights.
Crew transport	Two seven-seat crew vehicles docked at the despun node for the full campaign; one cargo port. Standard docking; no visiting vehicle matches a spin.
Radiation	Six months at 450 km and 51.6° is the ISS exposure regime. No shielding beyond standard module walls; tip water is not in a useful geometry and is not credited.

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9. Crew and campaign model

AGRE-F-007 asks for at least six months of exposure per subject and AGRE-F-014 for the study to be powered before it starts. Rev B baselines 180-day campaigns with ten crew: four at 1g, two at Mars gravity, two at lunar gravity, two in the rotating hub. The first campaign is a 60-day shakedown.

Result	Closure
Plant, microbial, fluids, thermal, mechanical, boundary	Within one campaign
Human dose response per band	Program result: five campaigns over about three years give ten subjects per partial-gravity band, ten in the rotating hub, twenty at 1g
Spin-rate sensitivity	Not closed by a fixed-rate vehicle

Between campaigns: data review, protocol refinement, hardware updates and maintenance. The platform is a mission-duration validation vehicle, not a permanent settlement.

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10. Mass budget

Item	Mass	Basis
Tip modules, 1g × 3	51.0 t	14 t hull + 3 t outfit each
Mars module	17.0 t	Common module
Moon module	17.0 t	Common module
Rotating hub module	14.0 t	Node-class, three radial ports
Arm trusses × 3	31.0 t	344 m at 30 kg/m
Arm utilities × 3: rail, power, fluid lines, MMOD	6.2 t	6 kg/m
Module radiator panels × 6	3.0 t	0.5 t each
Transit cars × 3	6.0 t	2 t each, two crew, pressurized
Tip trim tanks × 3, dry, and thruster pods × 3	2.7 t	
Despun node, pressurized	8.0 t	Three docking ports
Bearing, rotary transformer, optical link, drive	3.0 t	
The crossing: drum, three chambers, mechanisms	13.8 t	Crossing sizing, option C; replaces the 2 t transfer-lock placeholder
Solar arrays × 6, gimbals × 2, battery	6.0 t	0.4 t per 20 kW array; 1.2 t per gimbal; 82 kWh
Reboost propulsion dry, comms, GN&C, avionics	1.8 t	
Dry hardware	**180.5 t**	Rev B: 167.9 t
Growth, 20 % on dry hardware	36.1 t	
Trim and reserve water	12.0 t	Three 4 t tip tanks
Consumables at campaign start	10.0 t	7.2 t plus spares
Propellant	3.0 t	Spin-up reserve plus two years of reboost
On orbit, campaign start	**~242 t**	Excludes docked visiting vehicles. Rev B: ~227 t

Rev A carried no mass budget. Its "does not assume" list excluded a 250–500 t vehicle; Rev B landed under that, and Rev C does with the crossing in, because the arms are light and the modules are few, not because the vehicle is small. The dynamics model carries 289 t: this budget with the crew, the two docked crew vehicles, the cars' balance weights and the mass placed at two tips to make the arms even.

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11. Cost class

Rev A gave $600 M to $1.4 B, which is below one crew-rated module program with its launch. Rev B states a rough order of magnitude by element from ISS-module, Gateway-element and commercial-station analogs, with 30 % for program management and reserves. It is a cost class, not an estimate, and it is not a bid.

Element	Low	High
Common habitat module, non-recurring plus five units	$1.6 B	$2.4 B
Rotating hub module	$0.3 B	$0.5 B
Despun node, boundary hardware, arrays	$0.5 B	$0.9 B
Arms, cars, trim system, arm utilities	$0.3 B	$0.6 B
Integration, test, ground segment, operations development	$0.5 B	$0.8 B
Launch of ~240 t	$0.4 B, heavy lift, 3–4 flights	$2.0 B, medium lift, ~15 flights
On-orbit assembly campaign, robotic and EVA	$0.4 B	$0.7 B
Subtotal	$4.0 B	$7.9 B
To first campaign, with 30 % reserves	**$5.2 B**	**$10.3 B**
Operations per year: two crew flights, three cargo flights, ground	$0.8 B	$1.1 B

Development timeline of 8–10 years to first campaign is retained from Rev A and is consistent with the Gateway analog. The number that moved is the price. The crossing drum sits inside the boundary-hardware line and is not estimated separately; the Rev C mass moves the launch line by less than its own spread.

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12. Requirement closure

AGRE	Closure on Gradient One Rev C
F-001, F-007, F-008	Four gravity levels in one vehicle including a 1g control and a rotating 0g control sharing every confound; six-month exposures per subject
F-002 to F-006	Coriolis, cross-coupled head movement, inter-deck gradient, adaptation and inter-band transitions at 1.6 rpm; spin-rate sensitivity not resolved
F-009	Spin-up and spin-down with crew aboard are affordable and planned transients, including one simulated unplanned spin-down
F-010, F-011	Growth racks in every band against the 1g laboratory; arm lines and laboratory as the radial fluids test bed at 1.8 MPa head
F-012	Balance authority, wobble damping, car-transit excursions, and the hub/rotor boundary under rotation
F-013 to F-016	Campaign design and reporting; the human result is stated as a five-campaign program result

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13. What changed from Rev A

Item	Rev A	Rev B, and Rev C where marked
Rotor	Single truss, water counterweight "beyond the outer band"	Three arms at 120°, three identical 1g tip modules, self-balancing, major-axis spin
Hub	Non-rotating hub; docking and refuge; crossing unspecified	Rotating 0g hub on the rotor; small despun docking node; cyclic transfer lock, contactless power and data
0g control	In the non-rotating hub	Rotating hub, so rotation and gravity level are separable
Modules	"Pods", unsized	One common 17 t module, five copies, hung radially, three decks
Radial transit	Pressurized transition corridor	Pressurized car on an arm rail; spoke tube deferred, +60 t if adopted
Campaign	30–90 days, 2–4 crew	180 days, 10 crew, 60-day shakedown first
Mass	Not stated	~227 t bottom-up with 20 % growth; ~242 t in Rev C with the crossing
Cost	$0.6–1.4 B	$5–10 B to first campaign; ~$1 B per year to operate
Attitude, orbit, power, reboost, loads	Not stated	Stated above
The crossing (Rev C, November 2026)	None	Rev B's transfer lock, a 2 t placeholder, becomes the station's crossing at three chambers, 13.8 t, sized, modelled and flown in the flight software; the hub's dimensions fixed with it

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14. Open items

- **Coupled attitude simulation.** Done in the dynamics model (November 2026); despun node pointing and array shadowing remain.
- **The crossing.** Taken in Rev C. What remains: the leak budget per cycle, the chambers' outfit, the lock's tolerance on rate set with the bearing's real friction and the lock's real engagement, and a drum that coasts when its drive is lost.
- **The trim's law on the Rev C vehicle.** The drum's mass up the axis joins the node's in what a tip load tilts the principal axis against: a tonne at a tip wobbles the vehicle at 5.5×10^{-5} rad/s against Rev B's 3.0×10^{-5}, and the pods fire 80 s against 9 s to take it out. The wobble floor and the law's gain were sized on Rev B.
- **Arm first modes.** Stiffness and damping of a 344 m open truss with a moving 2 t car; whether the car's Coriolis load sets the rail design.

- **Module thermal at 1g.** Natural convection inside the 1g and partial-g modules changes cabin air handling and condensation behaviour from the ISS design basis.
- **Assembly sequence.** Arm erection and module hanging before spin-up, with the robotic and EVA campaign, is costed but not sequenced.

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15. Ownership and partnership

Developed under Aegis Station Infrastructure LLC. Engagement models: cooperative agreements, agency campaign partnerships, international research participation, commercial biological payload contracts. Full title remains with ASI unless otherwise negotiated. The Rev A and Rev B dossiers (September 2026) are retained as archives.
